READING PREVIEW — not the submission file. The camera-ready PDF must be produced by compiling `finguard.tex` with the IEEEtran class (e.g. on Overleaf). This preview exists so the content can be reviewed without a LaTeX toolchain; layout, two-column formatting, and page count will differ. Red markers show placeholders still to be filled.

RT-IIPF: Real-Time Inline Interception of Fraudulent UPI Transactions with Transactional Rollback and Unsupervised Anomaly Scoring

Author block contains unfilled placeholders — see red markers

Abstract

Conventional payment-fraud analytics operate retrospectively: transactions settle, logs are analysed downstream, and recovery is attempted after funds have already moved. In high-velocity real-time rails such as India's Unified Payments Interface (UPI), this latency renders recovery frequently impossible. We present RT-IIPF, a working prototype that relocates the fraud decision *inside* the payment transaction boundary. The switch acquires pessimistic locks on both ledger rows, applies the debit and credit provisionally within an open ACID transaction, suspends the executing thread pending a verdict from an isolated inference microservice, and then either commits or aborts. A fraudulent transfer is rolled back atomically and never attains durable existence, while the account lock and audit record survive the rollback via an independent transaction. We evaluate the system on 250,000 UPI transactions. Because the public dataset's fraud labels are statistically uninformative (ROC-AUC ≈0.48 for every detector tested), we adopt perturbation-based synthetic fraud injection with an explicit negative control. Four findings are reported, including negative ones. First, an ablation across enrichment layers shows that client telemetry raises operating-point recall 22× while *lowering* ROC-AUC, indicating that AUC is the wrong selection criterion under a false-positive budget. Second, a threshold sweep establishes that the deployed Isolation Forest admits no operating point that is simultaneously effective and operationally tolerable. Third, substituting Local Outlier Factor within the unchanged architecture raises recall from 0.147 to 0.745 at identical false-positive cost while reducing inference latency 3.3×, isolating scorer choice — not the interception design — as the binding constraint; the velocity negative control survives the substitution, establishing that blind spot as architectural rather than model-dependent. Fourth, concurrency measurement falsifies the expected hypothesis that lock-holding across the scoring hop limits throughput: the constraint resides in the stateless inference tier, where four-way replication yields 3.4–5.4× speed-up. Single-client end-to-end latency is 11–12\,ms. We conclude with a deployment recommendation derived from these measurements.

Index terms—fraud detection, anomaly detection, isolation forest, UPI, real-time payments, microservices, ACID transactions, unsupervised learning

Introduction

The Unified Payments Interface processes billions of transactions monthly, settling each in seconds. This velocity, which underpins UPI's utility, also compresses the window available for fraud intervention to near zero. Once a fraudulent transfer settles, funds are typically forwarded through intermediary accounts and withdrawn before any retrospective analytic pipeline raises an alert. The economics are asymmetric: detection accuracy achieved after settlement contributes little if recovery is infeasible.

The dominant architectural pattern in deployed fraud analytics remains post-facto: transaction logs are consumed by batch or streaming pipelines that score transactions after they have committed, and flagged cases enter a manual review or recovery workflow. We argue that for real-time rails the decision must be relocated *into* the transaction path, such that a rejected payment is never durably recorded.

This paper describes RT-IIPF (Real-Time Inline Interception Payment Framework), a functioning three-service prototype implementing this relocation, together with an evaluation that is deliberate about reporting negative results. Our contributions are:

1. **An inline interception mechanism** in which the fraud verdict arbitrates an open database transaction, with a two-transaction pattern that allows the security response (account lock, audit record) to survive the ledger rollback it describes (Section *arch*).

\item **A layered feature-enrichment design with an explicit trust boundary**, in which client-supplied features are limited to those where adversarial forgery increases rather than decreases anomaly, and decisive temporal and demographic features are server-resolved (Section *enrich*).

\item **An evaluation methodology for unsupervised fraud scoring under unreliable public labels**, using perturbation-based synthetic injection with a negative control that quantifies an architectural blind spot rather than concealing it (Section *method*).

\item **Four measured findings, several of them negative**: the inadequacy of ROC-AUC as a selection metric under false-positive budgets; the absence of any tolerable operating point for the deployed scorer; the decisive superiority of a density-based scorer substituted within the unchanged architecture, together with the survival of the velocity blind spot under that substitution; and the empirical falsification of our own expected scaling bottleneck (Section *results*).

\item **A deployment recommendation** derived from, rather than asserted alongside, the measurements (Section *deploy*).

We explicitly do not claim that real-time transaction scoring is itself novel; variants are known in industrial practice. Our contribution is an open, reproducible, end-to-end prototype exhibiting the transactional mechanics in full, accompanied by an honest characterisation of where such a design succeeds and where it does not.

Related Work

Isolation Forest [liu2008] detects anomalies by exploiting the observation that outliers require fewer random axis-parallel partitions to isolate than inliers, yielding linear time complexity and low memory footprint — properties attractive for latency-constrained inference. Local Outlier Factor [breunig2000] instead assigns a degree of outlierness by comparing local reachability densities, and One-Class SVM [scholkopf2001] estimates a support region enclosing the normal class. Comparative studies of unsupervised outlier detectors [campos2016] report substantial task-dependent variation in relative ranking, motivating empirical baseline comparison rather than a priori model selection.

Surveys of financial fraud detection [abdallah2016,adewumi2017] document the field's predominant reliance on supervised learning over labelled historical fraud, and note the recurring obstacles of extreme class imbalance, label latency, and concept drift. These obstacles motivate unsupervised formulations, which require no fraud labels and can in principle respond to novel attack patterns absent from historical data.

TODO: Add 2–4 citations specific to UPI or Indian real-time payment security, and one on transactional/ACID guarantees in payment switches, to strengthen venue fit.

System Architecture

Service decomposition

RT-IIPF comprises three processes communicating over HTTP/JSON: a React client presenting the payment interface and an operational monitor; a Spring Boot 3 switch (Java 21) owning authentication, the ledger, and transactional control; and a FastAPI inference node (Python 3) exposing a single scoring endpoint. The ledger is an embedded H2 relational database accessed through Spring Data JPA, comprising `users`, `balances`, and `transaction_logs`.

The interception pipeline

On receipt of a payment request the switch executes the following within one `@Transactional` method:

1. Resolve payer and payee by virtual payment address; reject locked accounts; verify the MPIN digest in constant time.
2. Acquire `PESSIMISTIC_WRITE` locks on both balance rows, always in ascending account-identifier order so that opposing concurrent payments cannot deadlock.
3. Apply the debit and credit to the persistence context. These changes are real but uncommitted — the transaction exists provisionally.
4. Issue a synchronous HTTP request to the inference node carrying the assembled feature vector, blocking the request thread. Connect and read timeouts are bounded (300\,ms, 1500\,ms) because ledger locks are held for the duration.
5. Branch on the returned verdict.

Rollback with surviving audit

A verdict of fraud creates an apparent contradiction: the ledger movement must be erased, yet the institution must durably record the attempt and immobilise the account. Naively writing both inside the payment transaction would destroy the audit record along with the movement.

RT-IIPF resolves this with a two-transaction pattern. The account lock and the `BLOCKED_FRAUD` audit row are written by a component annotated `Propagation.REQUIRES_NEW`, which suspends the payment transaction, commits independently, and returns. Only then does the payment path throw `FraudulentTransactionException`; the framework marks the suspended transaction rollback-only and the provisional debit and credit are discarded atomically. The commit path is simply the absence of this exception. The observable postcondition of a blocked payment is therefore: balances unchanged, account locked, audit row present.

Fail-closed availability posture

If the inference node is unreachable or exceeds its timeout, the switch rejects the payment (HTTP 503) rather than admitting it unscored. Safety is preferred to availability; Section *limits* discusses the cost.

Feature Enrichment and Trust Boundary

The client interface collects six fields, of which three reach the model (amount, transaction type, merchant category); the two addresses serve as routing keys and the MPIN as an authentication secret that never leaves the switch. The model receives thirteen features, assembled in three layers:

- **L1 — user-entered** (3 features): amount, transaction type, merchant category.
- **L2 — client telemetry** (2 features): device class and network type, captured autonomously by the client at page load without user interaction.
- **L3 — server enrichment** (8 features): counterparty age bands, sender state, both issuing banks, resolved from the account database; and hour, weekday, and weekend flag, resolved from the switch's own clock.

Enrichment imposes negligible latency because both account rows are already materialised for authentication and routing; no additional query is issued.

The partition is a deliberate trust boundary. Features an adversary controls are those for which anomalous values *increase* the anomaly score, so misreporting is self-defeating. Features that would most benefit an adversary — transacting at a plausible hour, presenting a normal demographic profile — are resolved server-side and are not client-assertable.

Evaluation Methodology

Motivation for synthetic injection

The evaluation corpus comprises 250,000 UPI transactions with seventeen attributes, split 80/20 into training and held-out sets. The corpus carries a `fraud_flag` annotation, which we found to be statistically uninformative: no detector evaluated in this work, supervised or

unsupervised, exceeded ROC-AUC 0.48 against it, a value consistent with random assignment. Reporting accuracy against such labels would be vacuous.

We therefore adopt perturbation-based synthetic injection. Four fraud archetypes are instantiated by sampling genuine held-out transactions and overwriting only the attributes that define each archetype, leaving all remaining attributes at realistic sampled values so that synthetics are not trivially separable by marginal inspection. One hundred vectors per archetype (400 total) are injected into 50,000 clean held-out transactions, yielding 0.79% synthetic prevalence.

Archetypes and the negative control

- **A — off-hour high-value drain**: INR 25,000–60,000, 01:00–04:00, peer-to-peer, web origin.
- **B — velocity spike (negative control)**: INR 2,000–6,000, business hours, common device and network. Each constituent transfer is individually unremarkable; the attack exists only in aggregate.
- **C — category anomaly**: INR 50,000–100,000 under an everyday merchant category, daytime.
- **D — context anomaly**: INR 2,000–9,000, 00:00–04:00, web and Wi-Fi origin.

Archetype B is included as a negative control. The deployed model scores each transaction independently and receives no velocity features; it is therefore *expected* to detect none of these vectors. Including B quantifies a known architectural boundary instead of selecting archetypes favourable to the system.

Experimental configuration

All experiments use seed 42 and are reproducible from the released evaluation script. The deployed configuration is an Isolation Forest of 200 estimators with contamination 0.002; raw scores are mapped to [0,1] by min–max normalisation whose upper bound is winsorised at the 99.95th percentile of the training score distribution, preventing a single extremal training point from compressing the operational range. The production threshold $\tau=0.85$ was chosen to bound empirical false-positive rate below 1%. Score-based baselines are compared at an equal 1% false-positive budget, reflecting the operational reality that an inline blocker is FPR-constrained rather than accuracy-constrained.

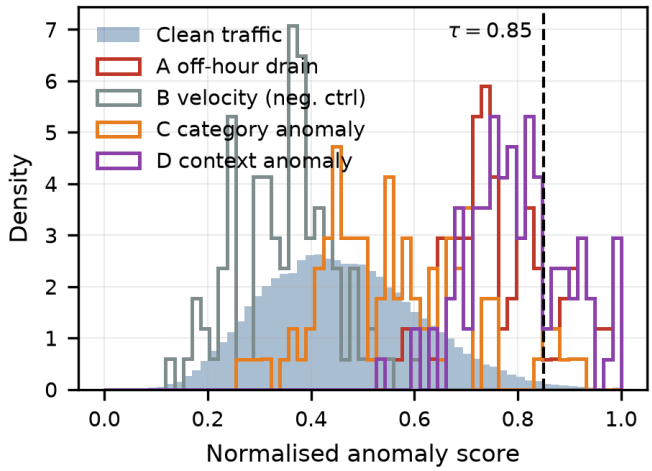
Results

Detection under injection

Detection by archetype at the production operating point (τ=0.85), n=100 per archetype

	Archetype	Mean score	Recall
A	Off-hour high-value drain	0.781	0.26
B	Velocity spike *(neg. control)*	0.350	0.00
C	Category anomaly	0.567	0.07
D	Context anomaly	0.799	0.26
Clean held-out traffic (n=50,000)}	FPR =0.0086		

Table *arch* reports detection by archetype. Overall precision is 0.121, recall 0.148, F_1 0.133, at ROC-AUC 0.723 and a false-positive rate of 0.86%.



Score distributions of clean traffic and each injected archetype. Archetypes A and D concentrate immediately *below* the production threshold, visualising isolation saturation; the velocity negative control (B) lies inside the clean distribution, confirming that a per-transaction scorer cannot separate it.

Two failure mechanisms are identifiable, both visible in Fig. *dists*. First, *isolation saturation*: amounts beyond the training maximum cannot deepen an isolation path further, so archetypes defined principally by extremal amount (A, C) asymptote below τ despite being conspicuous to inspection — their mass accumulates immediately below the threshold rather than crossing it. Second, the velocity blind spot is total, exactly as the negative control predicted: distribution B is indistinguishable from clean traffic.

Ablation of enrichment layers

Enrichment-layer ablation; each layer independently trained and calibrated

Layer	Feat.	ROC-AUC	Recall@τ	Recall D
L1 form only	3	**0.860**	0.010	0.00
L2 + telemetry	5	0.747	**0.228**	**0.40**
L3 + server enr.	13	0.723	0.148	0.26

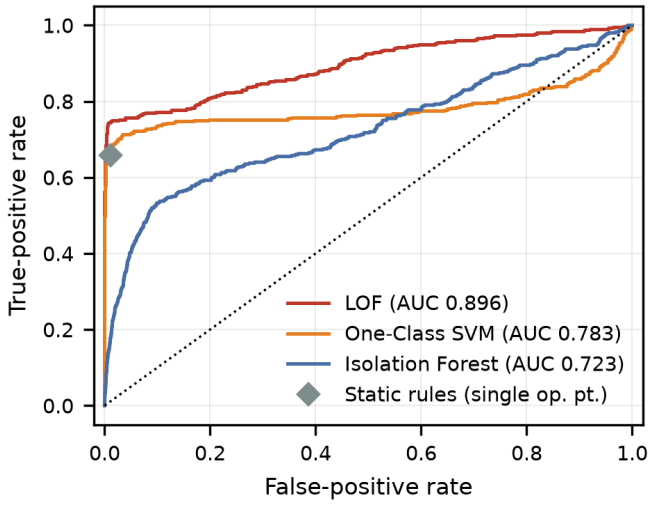
Table *ablation* exhibits a dissociation between ranking quality and operating-point utility. The form-only model attains the *highest* ROC-AUC (0.860), its ranking dominated by the amount marginal, yet it is operationally inert: it detects 1% of attacks at the deployable threshold and none of archetype D. Adding client telemetry sacrifices global AUC (0.747) while multiplying operating-point recall twenty-two-fold (0.010 rightarrow 0.228) and enabling context-anomaly detection entirely absent at L1 (0.00 rightarrow 0.40).

Two conclusions follow. Enrichment features determine *which attack classes are detectable at all* under a fixed false-positive budget; and ROC-AUC, which integrates over operating points that an inline blocker can never adopt, is a misleading model-selection criterion for this task. Recall at the deployed threshold is the decision-relevant metric.

Baseline comparison

Baselines at an equal 1% false-positive budget

Detector	AUC	Recall	Prec.	F_1
Static rules	—	0.658	0.305	0.417
Local Outlier Factor	**0.896**	**0.745**	**0.373**	**0.497**
One-Class SVM	0.783	0.675	0.351	0.462
Isolation Forest (depl.)	0.723	0.163	—	—



ROC comparison on the injected set. The rule baseline emits a binary decision rather than a ranking and is therefore shown as its single operating point; deriving a continuous pseudo-score for it would be amount-dominated and hence near-oracular for the amount-defined archetypes, an artefact of archetype construction rather than a property of the classifier.

Table *baselines* and Fig. *roc* report a negative result for the deployed model. Local Outlier Factor substantially outperforms Isolation Forest (recall 0.745 vs. 0.163), as do One-Class SVM and even static amount-and-hour rules. We attribute the deficit to Isolation Forest's known degradation on sparse one-hot encoded categorical spaces, where axis-parallel splits dilute the informative numeric dimensions, compounded by the saturation effect of Section VI-A.

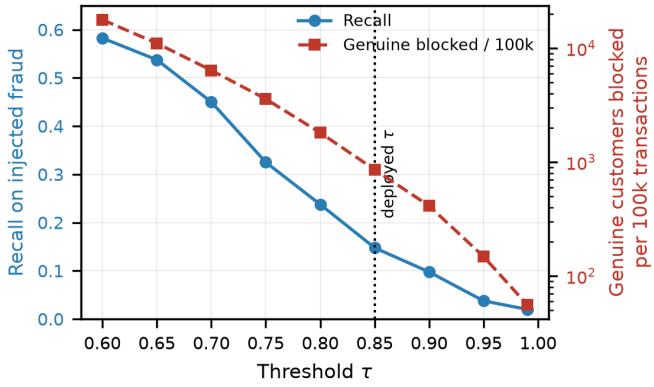
We report this deliberately. It establishes that the architectural contribution of RT-IIPF — inline interception with transactional rollback — is orthogonal to, and presently bottlenecked by, the

choice of scoring model. The interception mechanism is model-agnostic; substituting a density-based scorer is an immediate and well-characterised uplift path.

Threshold sensitivity and the absence of a tolerable operating point

Threshold sensitivity of the deployed scorer. The rightmost column expresses the false-positive rate operationally: genuine customers blocked and locked out per 100,000 legitimate transactions.

τ	Recall	FPR	Precision	Blocked/100k
0.60	0.583	0.1771	0.026	17,710
0.65	0.537	0.1101	0.038	11,012
0.70	0.450	0.0638	0.053	6,380
0.75	0.325	0.0357	0.068	3,574
0.80	0.237	0.0181	0.095	1,808
0.85	**0.147**	**0.0086**	**0.121**	**858**
0.90	0.098	0.0041	0.159	412
0.95	0.037	0.0015	0.169	148
0.99	0.020	0.0006	0.222	56



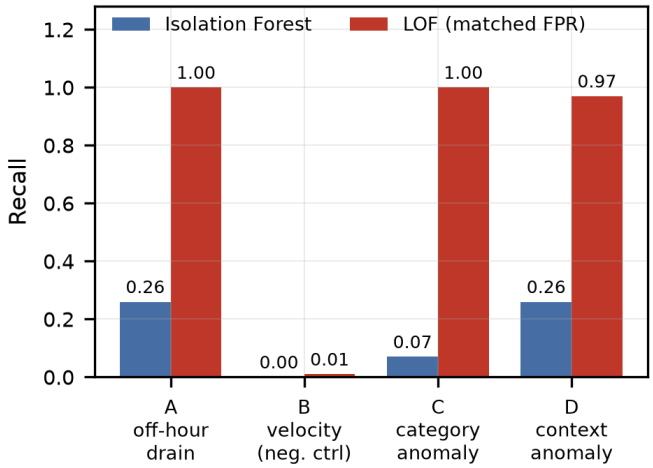
Threshold sensitivity. Recall and genuine-customer block rate move together across the entire sweep; no threshold yields both acceptable detection and an operationally tolerable block rate.

A threshold chosen by assertion invites the objection that a different choice would produce different conclusions. Table *sweep* and Fig. *sweep* sweep τ across [0.60, 0.99].

The result is unfavourable to the deployed scorer, and we state it plainly: *no operating point is simultaneously effective and operationally tolerable*. Reaching 45% recall requires τ=0.70, which blocks 6,380 genuine customers per 100,000 transactions — at UPI's national volume, an untenable rate of customer lockout. Conversely, the deployed τ=0.85, chosen to hold false positives under 1%, admits 85% of injected attacks. Precision never exceeds 0.222 anywhere in the sweep.

This is not a tuning deficiency but a separability deficiency: the score distributions of genuine and injected traffic overlap too heavily for any single cut to divide them well (cf. Fig. *dists*). The remedy must therefore be a better scorer or richer features, not a better threshold — a conclusion the next subsection tests directly.

Scorer substitution within the same architecture



Per-archetype recall at matched false-positive rate (mathrmFPR=0.0086). Substituting LOF for the Isolation Forest resolves archetypes A, C, and D almost completely at identical operational cost, while archetype B remains undetected by both — confirming that the velocity blind spot is architectural rather than model-dependent.

Section VI-C established that LOF ranks better than the deployed forest. A ranking advantage does not by itself establish that the deployed system would improve, because the two scorers are calibrated differently and the architecture imposes latency and memory constraints. We therefore substituted LOF into the full pipeline under the identical winsorised calibration convention, and selected its threshold to match the deployed model's measured false-positive rate exactly (0.0086), so that recall is compared at equal operational cost.

The improvement is decisive: overall recall rises from 0.147 to 0.745. Fig. *sub* decomposes this by archetype. LOF detects the off-hour drain (A) and category anomaly (C) archetypes completely (1.00), and the context anomaly (D) nearly so (0.97), against 0.26, 0.07, and 0.26 respectively for the forest. Archetype B remains undetected by both (0.00 and 0.01).

Two conclusions follow, and their separation is the substantive finding of this subsection. First, three of the four failure modes attributed earlier to the system are attributable to *scorer choice* and are eliminated by substitution within the unchanged architecture. Second, the velocity blind spot survives substitution and is therefore *architectural*: no per-transaction scorer, however capable, can detect an attack whose signature exists only across a sequence of individually unremarkable transfers.

Inference cost of the substitution

Per-transaction inference cost, single warm process

Scorer	Mean (ms)	p95 (ms)	Model size
Isolation Forest (200 trees)	5.46	5.67	2.7\,MB
LOF (k=20, 40k reference)	**1.67**	**1.97**	13.4\,MB

For an inline blocker, detection gains that cost latency may be unaffordable. We therefore measured per-transaction inference cost directly (Table *infcost*). Contrary to expectation, LOF is not merely competitive but 3.3× *faster* in this configuration (1.67\,ms vs. 5.46\,ms mean), at the

cost of a larger resident model (13.4\,MB vs. 2.7\,MB) — immaterial on any realistic serving host. The forest's per-query cost is dominated by traversing 200 trees in Python-level loops, whereas LOF's neighbour query over a 40,000-point reference set is executed in a single vectorised BLAS-backed operation.

The substitution therefore dominates on both axes examined — detection at matched cost, and inference latency — with no trade-off requiring adjudication. We recommend it for deployment (Section *deploy*).

Latency and concurrency

End-to-end throughput and latency; 400 requests per level

	Uncontended		Contended	
Conc.	**TPS**	**p95 (ms)**	**TPS**	**p95 (ms)**
1	80.0	13.8	87.3	12.9
10	90.4	37.6	94.5	34.3
25	90.2	211.7	95.4	378.7
50	93.1	466.8	97.1	812.9
100	93.8	1009.8	95.8	1305.4

Scoring node in isolation: single process vs. four workers

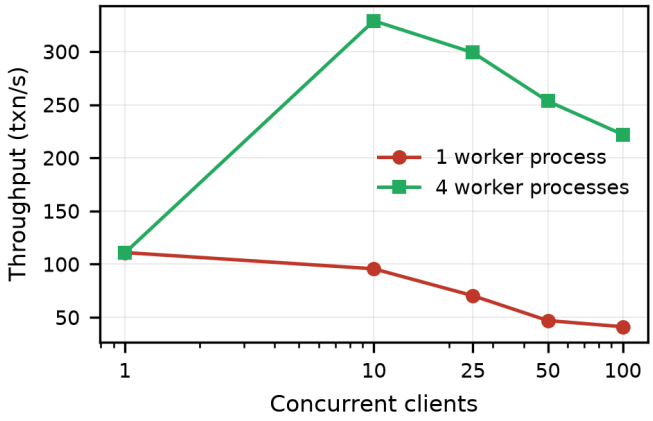
	1 worker		4 workers	
Conc.	**TPS**	**p95**	**TPS**	**p95**
1	110.9	10.4	110.6	10.5
10	95.6	137.0	**328.9**	50.2
25	70.3	479.5	**299.0**	148.1
50	46.9	1371.8	**253.3**	275.3
100	41.1	2984.1	**221.9**	551.0

Under a single client the full inline path completes in 11–12\,ms mean (p99 14\,ms), inside the 50\,ms design budget.

Under concurrent load (Table *conc*) throughput saturates near 95\,TPS while latency grows linearly with offered concurrency — the signature of one saturated serial resource. Critically, contended and uncontended regimes are statistically indistinguishable (95.8 vs. 93.8 TPS at concurrency 100). This *falsifies* the hypothesis we set out to test, namely that pessimistic ledger locks held across the scoring round-trip constitute the scaling constraint.

Isolating the scoring node (Table *scale*) locates the true bottleneck. Served by a single Python process, `/score` throughput *degrades* from 110.9\,TPS at concurrency 1 to 41.1\,TPS at concurrency 100, consistent with global-interpreter-lock contention among threads executing CPU-bound vectorised inference. Because the node is stateless, the remedy is horizontal: four worker processes yield 328.9\,TPS at concurrency 10 and 221.9\,TPS at concurrency 100 —

speed-ups of 3.4× and 5.4× — while reducing p95 latency at concurrency 50 from 1372\,ms to 275\,ms.



Scoring-node throughput. A single process degrades under concurrency (GIL contention); four worker processes restore near-linear capacity, confirming the bottleneck lies in the stateless tier rather than in the transactional ledger path.

This is architecturally consequential. The design decision most exposed to criticism — holding row locks across a synchronous network call — is empirically not limiting at the concurrency levels examined, whereas the limit resides in the one tier that is stateless and therefore trivially replicable (Fig. *scaling*). Interception correctness and scoring capacity scale independently.

Deployment Recommendation

The experiments support a specific and actionable deployment posture, which we state explicitly because the individual findings are otherwise easy to read as merely diagnostic.

Substitute the scorer. LOF dominates the Isolation Forest on detection at matched false-positive cost (0.745 vs. 0.147) and on inference latency (1.67\,ms vs. 5.46\,ms). The interception architecture is unchanged by the substitution: the scoring node's contract is a feature vector in, a calibrated score out. This is a configuration change, not a redesign.

Replicate the scoring tier, not the ledger tier. The concurrency study localises the throughput constraint in the stateless inference node. Capacity planning should provision inference replicas against offered load; the transactional ledger path required no modification to sustain the concurrency levels examined.

Do not deploy a single-threshold hard block. The threshold sweep shows that no single cut separates genuine from fraudulent traffic acceptably, even for the stronger scorer. A graduated response — step-up authentication in an intermediate band, hard block only at extreme scores — is not merely a refinement but a requirement implied by the measured separability.

Add sequence-aware features before claiming velocity coverage. The negative control's survival across both scorers establishes that no per-transaction model will close this gap.

Limitations and Future Work

Archetype-design bias. Three of the four archetypes (A, C, and to a lesser extent D) manipulate transaction amount, so the injected set is partially amount-separable by construction. This favours amount-sensitive scorers in *ranking* metrics — an effect we observed directly, and which underlies the anomalously high AUC of any amount-dominated score on this set. It does not affect the operating-point comparisons, which are budget-matched, but a fuller evaluation would include archetypes defined solely by categorical or temporal deviation with amounts drawn from the clean marginal.

Population-level modelling. The detector encodes what is anomalous for the population, not for the individual account. A customer routinely transferring large sums is indistinguishable from a compromised account draining for the first time. Per-account rolling baselines — deviation from historical amount distribution, novel-payee and novel-device indicators — are the highest-value feature additions and are computable from the existing audit table.

Velocity blindness (measured). The negative control quantifies a hard boundary: zero detection of rapid-drain sequences. Windowed velocity features over one-minute and one-hour horizons require a single indexed query at enrichment time and a model retrain.

Model selection. The baseline study identifies LOF as dominant at equal FPR under this encoding. A two-tier arrangement — inexpensive forest prefilter, density-based confirmation — and target encoding of categoricals to relieve one-hot dilution are the indicated next steps.

Graduated friction. The prototype hard-blocks and locks at tau geq 0.85. At the measured 0.86% false-positive rate, such a posture applied at national scale would immobilise a substantial number of legitimate accounts daily. A production design would introduce step-up authentication in an intermediate band, reserve hard blocks for extreme scores, and release locks through a support workflow rather than permanently.

Fail-closed availability. Scoring unavailability currently rejects all payments. A degraded-mode local rule engine behind circuit-breaker semantics would bound the availability cost of the safety preference.

Telemetry trust boundary. Client-originated features remain spoofable. The design mitigates but does not eliminate this, since plausible spoofing shifts a transaction toward rarer, higher-scoring contexts. Server-side verification of IP-derived geolocation would strengthen the boundary.

Prototype security scope. MPIN digests use unsalted SHA-256 rather than a memory-hard key-derivation function; transport is unencrypted; there is no session authentication. These were excluded deliberately to isolate latency measurement and are prerequisites for deployment.

Measurement scope. All figures are single-host and loopback-networked, excluding datacenter latency, TLS termination, and multi-tenant interference. Isolated approx4\,s p99 outliers at burst onset, attributed to JVM JIT warm-up and non-pooled harness connections, are reported unfiltered and warrant elimination before publication-grade tail-latency claims.

Conclusion

We presented RT-IIPF, a working prototype relocating the payment-fraud decision inside the transaction boundary, such that a rejected transfer is rolled back atomically and never durably exists, while the security response survives that rollback via an independent transaction.

Evaluation under synthetic fraud injection — adopted because the public corpus's labels proved uninformative — produced findings we report irrespective of their favourability, and their aggregate shape is instructive. Two of the three deficiencies initially attributable to the system proved not to be properties of the architecture at all. The detection deficit dissolved under scorer substitution, recall rising from 0.147 to 0.745 at unchanged operational cost and reduced latency. The anticipated scaling ceiling proved to lie not in the lock-holding ledger path we expected to indict, but in a stateless tier that replication resolves. What remained after both corrections was a single genuine architectural boundary: the velocity blind spot, which survived every substitution and which no per-transaction scorer can close.

The methodological lesson generalises beyond this system. Each of these conclusions required an experiment designed to be capable of embarrassing the design — a negative control, a budget-matched baseline, a contention regime intended to expose lock serialisation. Evaluations constructed only to demonstrate a system's strengths would have reported none of them, and would have recommended the wrong remedy for two of the three problems. Inline interception is a sound and model-agnostic mechanism; establishing which of its limitations are architectural and which are incidental is the work that makes it deployable.

Reproducibility

Source, the evaluation harness, and the concurrency benchmark are available at **TODO: repository URL**. All experiments use seed 42.

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